# AGREEMENT ANALYZER

*Helping a borrower understand a loan agreement before signing it*

**The problem.** Instant-loan applications hand first-time borrowers long contracts in legal English. The terms that decide the real cost — a fee deducted before disbursal, a rate the lender may revise at will, a penalty charged per month, permission to telephone the borrower's family — sit in the fine print.

**What it does.** Upload a PDF, a scan, a photograph or a Word file. The system identifies the kind of agreement, marks every risky clause line by line on the original page, explains it in English or Hindi with speech output, computes the true annual cost, and answers the borrower's own questions.

**Why it is not simply a chatbot.** A general assistant answers from its training data and may describe a clause the contract does not contain. Here the document itself is the knowledge base: passages are embedded locally as 384-dimension vectors and retrieved by similarity, then combined with twenty-seven rules that read obligation and negation. Nothing is sent away to be searched, and the document, its analysis and its index are all encrypted.

**Measured, not claimed.** Document type is classified at **91.7 per cent** under cross-validation. On the same forty labelled clauses the rules judge harm correctly **80.0 per cent** of the time against **45.0 per cent** for a learned classifier — which is why rules decide risk and embeddings only retrieve. **Thirty-seven automated tests** run on every change.

**In one line:** a loan advertised at 24 per cent really costs **42.8 per cent** once a single ₹4,500 fee is counted.